

PROJECT AND TEAM INFORMATION

JEEVANROUTE: Emergency ambulance selection, hospital capability matching and route calculation

Student / Team Information

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PROPOSAL DESCRIPTION

Motivation (1 pt)

- 1. I notice that in areas emergency medical care often faces delays. Healthcare facilities are scarce ambulances are hard to find connectivity is poor. It is difficult to locate the correct hospital.
- 2. When a situation is critical I realize that choosing the hospital may not solve the problem. The patient may need ICU beds, CT scans, specialists, blood or urgent treatment. These delays can greatly reduce the chance that the patient receives care.
- 3. The idea behind JEEVANROUTE is derived by the need of emergency ambulance and hospital in times of emergency and reduce delays . It will evaluate the emergency level of patient and select appropriate ambulance and hospital based on hospital and severity level .
- 4. JEEVANROUTE is an application that combines DATA STRUCTURE AND OBJECT ORIENTED PROGRAMMING in CPP . DS concepts like priority queues , graphs and lists . It also uses OOPS concepts like Classes , objects , encapsulation , inheritance and polymorphism to solve real world problems . It is Useful and gives experience of real world problem .

State of the Art / Current solution

- 1. Patient assessment: It tracks the location of patient and what severity they are facing.
- 2. Emergency Prioritization : JEEVANROUTE prioritizes patient based on their severity level . It uses priority queue so that emergency cases are prioritized.
- 3. Ambulance Management : JEEVANROUTE software tracks the location and availability of ambulance and its condition before departing it .
- 4. Hospital Selection: JEEVANROUTE select's hospitals based on their quality and availability of the services like ICU beds CT scans, doctors and specialist required in certain patient and availability of blood.
- 5. Route Planning: JEEVANROUTE represents the road network as a graph enabling the calculation of the route for the ambulance.
- 6. Ambulance Dispatch: JEEVANROUTE sends the ambulance to the patient's location after selecting the ambulance and hospital.
- 7. Hospital Pre-alert: JEEVANROUTE sends details about the patient and emergency condition to the chosen hospital allowing staff to prepare before the ambulance arrives.
- 8. Data Management: JEEVANROUTE stores patient, ambulance, hospital and route information in data structures.
- 9. OOP Implementation JEEVANROUTE uses C++ classes and objects applying concepts such, as encapsulation, inheritance and polymorphism.

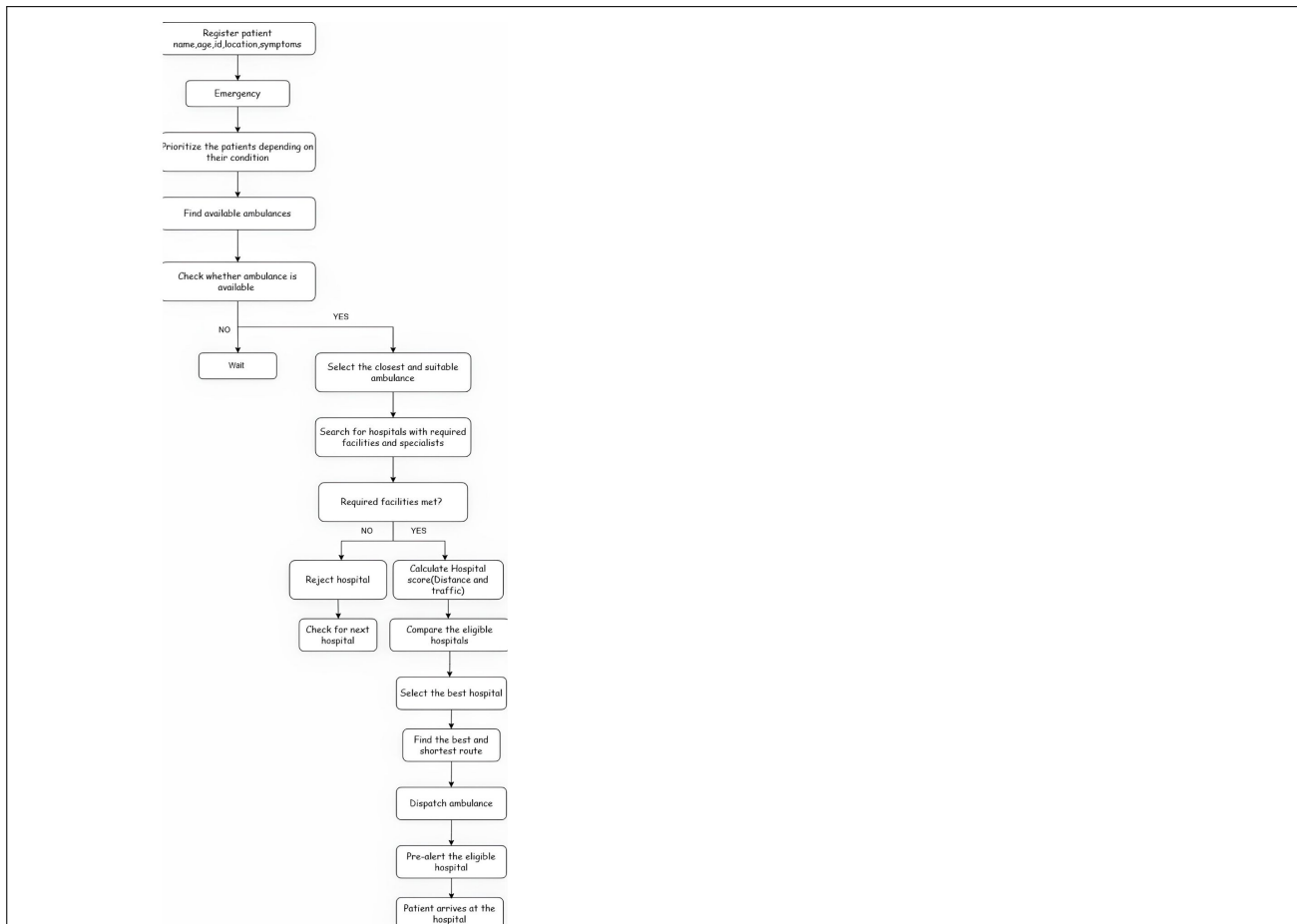
Project Goals and Milestones

- * Reduce Emergency Response Time: Aim to shorten the time it takes to get to patients and bring them to hospitals. Faster response means chances for recovery.*
- * Prioritize Critical Patients: Use a system that decides which patients need help first. The serious the condition the faster they get attention.*
- * Manage Ambulances Efficiently: Keep a record of where each ambulance is. Know if it is available how people it can carry and what its current status is. This helps assign the ambulance to the right call.*
- * Select Suitable Hospitals: Choose hospitals that can handle the patient's condition. Look for ICU beds CT scanners, specialists and blood supplies. Make sure the hospital can actually treat the patient.*
- * Calculate Efficient Routes: Use a map as a network of roads. Find the path by checking distance and how long it takes to travel. The fastest route saves time.*
- * Improve Emergency Coordination: patients, doctors, ambulance drivers and hospitals in one shared system. Better communication means care.*
- * Provide Hospital Pre-alerts: Send a warning to the chosen hospital before the ambulance arrives. Let them know about the patient and the type of emergency. This gives the hospital time to prepare.*
- * Improve Resource Utilization: Match ambulances and hospital resources with what each patient requires. Avoid waste. Keep the system running smoothly.*

Project Approach

- * Patient Assessment: Patient Assessment gathers info, about the patient, where they're the type of emergency they face.*
- * Priority Assignment: Priority Assignment decides which emergencies are serious and places the match top of the list using a priority queue.*
- * Ambulance Selection: Ambulance Selection looks for the ambulance that fits the situation.*
- * Hospital Selection: Hospital Selection chooses a hospital that has the equipment and can handle the patient's needs.*
- * Route Calculation: Route Calculation uses graphs to find the way to reach the patient.*
- * Dispatch Pre-: Dispatch Pre- sends the ambulance to the patient. Informs the hospital that an emergency is arriving.*
- * System Integration: System Integration puts all parts of the system together using C++ data structures and object-oriented programming concepts.*

System Architecture (High Level Diagram)



Project Outcome / Deliverables

1. C++. Object-Oriented Programming concepts.
2. Data Structures concepts. Stacks, Queues, Priority Queues, Lists and Graphs.
3. Standard C++ Library documentation.
4. Algorithms, for path and graph traversal.
5. Articles related to ambulance routing and emergency healthcare systems. i). The Rwanda912 paper introduces a ****Destination Decision Support Algorithm (DDSA)****. This algorithm helps ambulance teams pick the hospital when there is an emergency. It looks at the patients condition. It also looks at where the emergency's happening. It checks where the hospital is located. It considers if the hospital is ready. It checks the staff available. It looks at the beds, in the hospital. It checks the equipment there. It also looks at the blood supply. It takes into account the treatments that are needed. The hospital that is suggested can then be. Changed by the ambulance staff or the dispatchers. ii). Almalki Aldhahri Aljojo (2025) introduced an ambulance routing system that uses real-time EMS availability. The system picks the routes and medical facilities by looking at the availability of ambulances and hospitals the distance involved and how quickly help can arrive. The main purpose is to cut down on delays and make sure patients get access, to the right kind of care. This aligns with JeevanRoute's method, which also combines distance, traffic conditions, hospital capabilities and real-time updates to improve response efficiency.

Assumptions

- * Patient and emergency details are entered correctly.*
- * I see that Ambulance location Ambulance availability and Ambulance status are updated regularly.*
- * I confirm that Hospital facilities and Hospital bed availability are accurate.*
- * I note that Road distances and Road routes are predefined.*
- * I can rely on GPS data and location data because they are available.*
- * I trust that suitable Ambulance and suitable Hospital are assumed to be available.*
- * I understand that the system supports. Does not replace, professional medical decisions.*

References

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5. *GeeksforGeeks — Array Data Structure — Used for implementing the spreadsheet grid. Site → geeksforgeeks.org/array-data-structure*
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